Question:
What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?
Answer Choices:
A. 2.48 × 10^-8 cm
B. 5.92 × 10^7 cm/sec
C. 8.66 × 10^-6 cm
D. 3.14 × 10^-5 cm
E. 9.81 × 10^-4 cm
F. 5.27 × 10^-9 m
G. 1.23 × 10^-7 cm
H. 4.14 × 10^-10 m
I. 6.625 × 10^-27 erg sec
J. 0.512 × 10^6eV

Reasoning + Answer:

<tool_call> Reasoning process here:

We are asked to find the de Broglie wavelength of an electron with a kinetic energy of 1 eV.

The de Broglie wavelength is given by:

Î» = h / p

where:
- h = Planck's constant = 6.626 ÃĹ 10âģ»Â³âģ´ JÂ·s
- p = momentum of the electron

For non-relativistic electrons, kinetic energy K = (1/2) mvÂ², and momentum p = mv.

We can relate kinetic energy to momentum via:

p = âĪļ(2mK)

So:

Î» = h / âĪļ(2mK)

We need to compute this for K = 1 eV = 1.602 ÃĹ 10âģ»Â¹âģ¹ J

Mass of electron m = 9.109 ÃĹ 10âģ»Â³Â¹ kg

Step-by-step:

1. Compute 2mK:
2 ÃĹ (9.109 ÃĹ 10âģ»Â³Â¹) ÃĹ (1.602 ÃĹ 10âģ»Â¹âģ¹) = 2 ÃĹ 9.109 ÃĹ 1.602 ÃĹ 10âģ»âģµâģ°
âīĪ 2 ÃĹ 14.59 ÃĹ 10âģ»âģµâģ° = 29.18 ÃĹ 10âģ»âģµâģ° = 2.918 ÃĹ 10âģ»âģ´âģ¹

2. Take square root:
âĪļ(2.918 ÃĹ 10âģ»âģ´âģ¹) âīĪ 5.40 ÃĹ 10âģ»Â²âģµ kgÂ·m/s

3. Compute Î» = h / p = (6.626 ÃĹ 10âģ»Â³âģ´) / (5.40 ÃĹ 10âģ»Â²âģµ)
âīĪ 1.227 ÃĹ 10âģ»âģ¹ m

Convert to cm: 1.227 ÃĹ 10âģ»âģ¹ m = 1.227 ÃĹ 10âģ»âģ· cm

So Î» âīĪ 1.23 ÃĹ 10âģ»âģ· cm

Now check the options:

A. 2.48 ÃĹ 10âģ»âģ¸ cm âĨĴ too small
B. 5.92 ÃĹ 10âģ· cm/sec âĨĴ units wrong (velocity)
C. 8.66 ÃĹ 10âģ»âģ¶ cm âĨĴ too large
D. 3.14 ÃĹ 10âģ»âģµ cm âĨĴ too large
E. 9.81 ÃĹ 10âģ»âģ´ cm âĨĴ too large
F. 5.27 ÃĹ 10âģ»âģ¹ m âĨĴ 5.27 ÃĹ 10âģ»Â¹Â¹ cm âĨĴ wrong value
G. 1.23 ÃĹ 10âģ»âģ· cm âĨĴ matches our calculation
H. 4.14 ÃĹ 10âģ»Â¹âģ° m âĨĴ 4.14 ÃĹ 10âģ»Â¹Â¹ cm âĨĴ wrong
I. 6.625 ÃĹ 10âģ»Â²âģ· ergÂ·s âĨĴ units wrong
J. 0.512 ÃĹ 10âģ¶ eV âĨĴ energy, not wavelength

So the correct answer is G: 1.23 ÃĹ 10âģ»âģ· cm

Note: Option G is written in cm, and our calculation gives 1.23 ÃĹ 10âģ»âģ· cm, which matches.

<answer> G </answer>